

dashboards to visualise your data in a variety of charts, tables and maps.

4) Ansible

Ansible is an open source IT automation tool. It provides configuration management, infrastructure orchestration, task automation and more advanced IT tasks such as continuous deployments or zero downtime rolling updates. Ansible's main feature is that it is agentless — no software or agent is installed on the client that communicates back to the server. Ansible is written in Python and uses YAML for playbook language, both of which are considered relatively easy to learn.

5) Kubernetes

Kubernetes is an open source container orchestration system designed for the automation of deployment and scaling, as well as the management of containerised applications. It was originally designed by Google and is now maintained by the Cloud Native Computing Foundation. Kubernetes ranks No. 8 among the top open source projects on GitHub, based on the contributor count.

6) Nagios

Nagios is one of the most popular free and open source DevOps monitoring tools. In case of failure, it can send alerts to the team, allowing it to fix the problem before outages affect business processes, end users or customers. Nagios stores the logs of events and failures, which helps to forecast outages and errors and detect security threats. It is flexible, supporting both agent-based and agentless architectures. Nagios has a huge community of supporters and contributors. You can use the existing thousands of plugins and add-ons. Nagios allows users to easily develop their own service checks depending on needs, by using their tools of choice (shell scripts, C++, Perl, Ruby, Python, PHP, C#, etc).

7) Docker

Docker is the leading open source container platform. It is designed to make it easier to create, deploy and run applications by using containers, which allow a developer to pack together an application with all of the parts it needs, such as source code and other dependencies. Multiple containers can run on the same machine and share the OS kernel with other containers, each running as isolated processes in the user space. Containers take up less space than VMs, can handle more applications and require fewer VMs and operating systems.

8) Gradle

Gradle is an open source build automation tool. It has been designed to support build automation across multiple languages and platforms including Java, C/C++, Python and Groovy, and it has support for IDEs and CI tools such as Eclipse, IntelliJ and Jenkins. Gradle build scripts are written using a Groovy or Kotlin DSL.

Features:

- Highly customisable — Gradle is modelled in a way that is customisable and extensible in the most fundamental ways.
- Fast — Gradle completes tasks quickly by reusing output from previous executions, processing only inputs that are changed and executing tasks in parallel.
- Powerful — Gradle is the official build tool for Android, and comes with support for many popular languages and technologies.

9) Selenium

Selenium is one of the most popular open source testing tools for Web applications. It was originally developed by Jason Huggins in 2004 for testing an internal project at ThoughtWorks.

Selenium supports automation testing of Web based applications and the largest browser vendors. It has five basic components — the Selenium IDE, Selenium Remote Control, Selenium WebDriver, Selenium Grid and the Selenium client API (introduced in Selenium 2). The Selenium IDE is the Firefox add-on for record-and-playback Web application tests. WebDriver directly communicates with the Web browser and uses its native compatibility to automate.

10) Puppet

Puppet is an open core software configuration management tool. One of its main features is that the infrastructure can be managed as a code. Similar to Ansible, Puppet automates infrastructure provisioning, which allows for faster and safer deployment of applications and software.

Features:

- Infrastructure automation
- Automated provisioning
- Application release automation
- Task management
- Visualisation and reporting
- Orchestration
- Code management
- Role based access control
- Discovery and insights

Apart from these top of the list open source DevOps tools, there are many other open source tools too that are good choices like Vagrant, Apache Ant, Maven, Chef, etc. **END** 

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